Correction to Wei, Nathoo, and Masson (2023, Appendix C)

2024-02-26

As one anonymous referee noted, "Figure 11: given that the posterior-mean distributions do not overlap for the *BayesFactor* and the *rmBayes* package, I wondered whether the two methods actually implement the same model and priors?"

We have since identified errors in the original simulation code detailed in Appendix C and the supplementary material at https://osf.io/x2pvw/, which involved *Monte Carlo error and a data permutation issue*. The '*BayesFactor*' **R** package uses a reduced parameterization approach to model fixed effects, as described on page 1762 of the manuscript. Conversely, our *rmHDI* function from the '*rmBayes*' **R** package defaults to treating condition effects as random effects (refer to the `treat` argument). This results in a difference in model specifications between the two packages.

We have corrected these discrepancies by updating our model to assume fixed effects, rerunning the simulations, and revising Figure 11 accordingly.

p. 1778

In Fig. 11, two density plots (permuted data but using a constant random seed, or not permuted but varying the random seed) generated from results provided by the rmHDI function in the '*rmBayes*' package highly overlap, indicating that permutation of the data produces variability in outcomes of a similar magnitude to setting different random seeds. Moreover, the functions in the '*BayesFactor*' package returned less variability in estimates for posterior means ~~but more variability in estimates for standard error of the mean (thus, interval width) and posterior mean difference, whereas the rmHDI performance is quite the opposite~~. The separation percentage is less variable when calling rmHDI, as shown in panel E of Fig. 11. Although the models and priors assumed by the two packages are identical, there may be differences in the actual code implementation, especially for Equations 2 and 4 and MCMC samplers (Gibbs sampling in the *anovaBF* and NUTS in the *rmHDI*), leading to differences in the variable results.

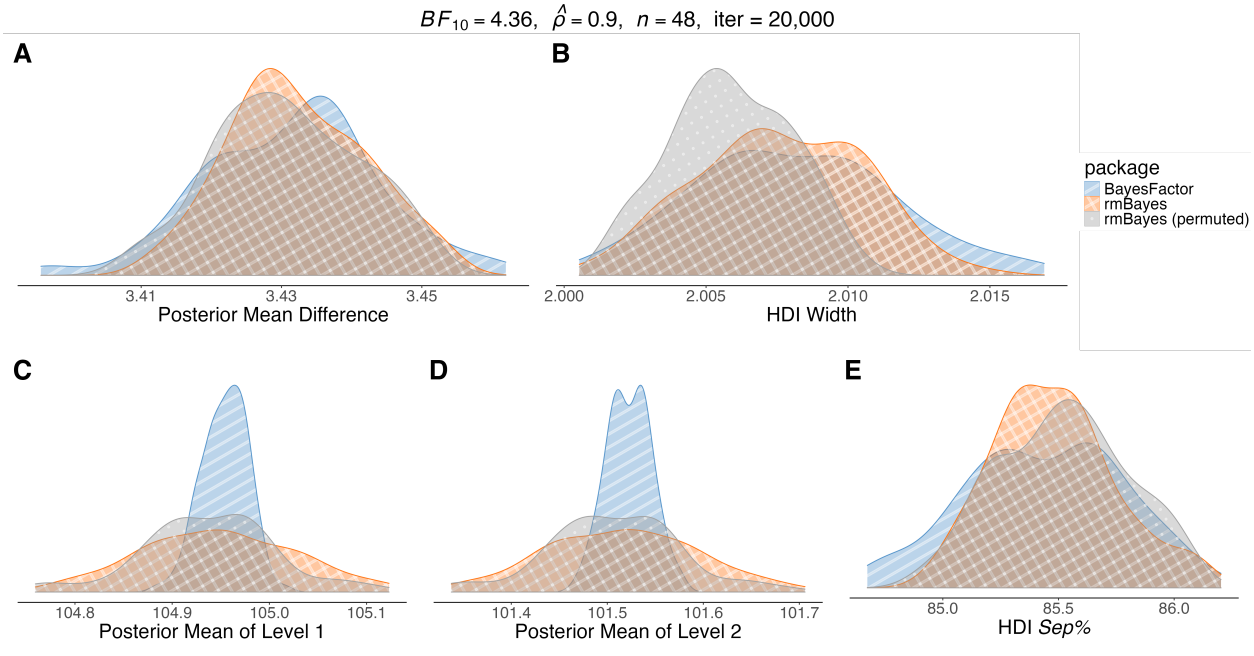


Fig. 11 Density plots of the simulations from replicating `R` functions. **A**: posterior mean difference; **B**: HDI width; **C**: posterior mean of one condition; **D**: posterior mean of the other condition; **E**: HDI separation percentage. The same random seed was fixed when investigating the permutation issue, denoted as '*rmBayes (permuted)*', whereas different random seeds were used for investigating the Monte Carlo error in '*BayesFactor*' and '*rmBayes*'

Wei, Z., Nathoo, F. S., & Masson, M. E. J. (2023). Investigating the relationship between the Bayes factor and the separation of credible intervals. *Psychon. Bull. Rev.*, *30*, 1759–1781. doi: 10.3758/s13423-023-02295-1